System Identification and Preliminary Control of a Propeller-Actuated Beam

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1 Introduction

The propeller-actuated beam (Fig. 1) rotates in the horizontal plane; its angular position θ is driven by propeller thrust applied at servo angle α. Accurate modeling of the coupled chain $u \to \alpha \to \theta$ is essential before closing the control loop.

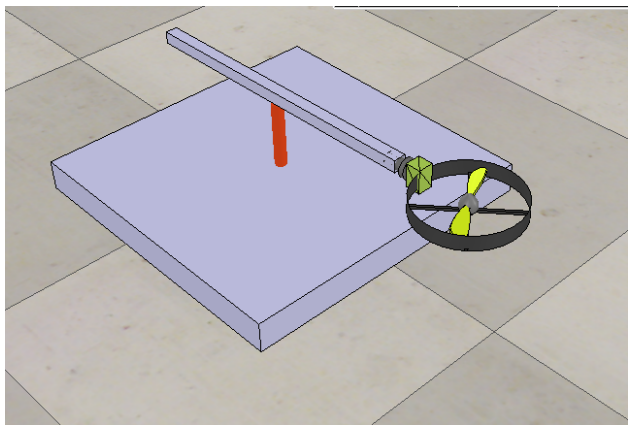

Figure 1: V-REP digital twin of the mechanism.

From first principle physical dynamics we get:

$$I_b \ddot{\theta} = F_T L_b \sin\alpha \approx F_T L_b \, \alpha \tag{1}$$

Once the system is linearized the transfer function is achieved:

$$\frac{\theta(s)}{\alpha(s)} = \frac{c}{s^2}, \qquad \frac{\alpha(s)}{u(s)} = \frac{b}{s+a}. \tag{2}$$

Unknowns a, b and c are identified in Parts A–C.

2 Experimental Methodology

Two CSV data sets were exported from V-REP:

1. **Experiment 1**: (u, α) for servo identification of parameters a and b, given sample time $h = 0.05\,\mathrm{s}$.

2. **Experiment 2**: (u, θ) for identification of parameter c given found values of a and b, with same sample time

Signals were converted to radians, zero-meaned, and purged of `NaN/Inf` rows at the start of data processing.

3 Part A – Servo Model $G_s(s)$

Summary:
This section models the servo dynamics relating input commands $u(k)$ to actual servo angles $\alpha(k)$. Ordinary Least Squares (OLS) regression is used to estimate the model parameters from V-REP simulation data. The identified model is essential for constructing the full beam response in later sections.

$$G_s(z) \;=\; \frac{b_1\, z^{-1}}{1 - a_1\, z^{-1}} \quad \Longrightarrow \quad \alpha(k) = a_1\, \alpha(k-1) + b_1\, u(k-1).$$

3.1 Ordinary Least Squares

Define

$$\mathbf{Y} = \begin{bmatrix} \alpha(1) \\ \alpha(2) \\ \vdots \\ \alpha(N) \end{bmatrix}, \qquad \mathbf{\Phi} = \begin{bmatrix} \alpha(0) & u(0) \\ \alpha(1) & u(1) \\ \vdots & \vdots \\ \alpha(N-1) & u(N-1) \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}.$$

The normal equations $\mathbf{\Phi}^\top \mathbf{\Phi}\, \hat{\boldsymbol{\theta}} = \mathbf{\Phi}^\top \mathbf{Y}$ yield the closed-form estimator

$$\boxed{\hat{\boldsymbol{\theta}} = (\mathbf{\Phi}^\top \mathbf{\Phi})^{-1}\, \mathbf{\Phi}^\top \mathbf{Y}} \tag{A1}$$

Using samples from `Experiment1` with $h = 0.05\,\mathrm{s}$,

$$\hat{a}_1 = 0.4188, \quad \hat{b}_1 = 0.5298,$$

3.2 ZOH Discretization Approach

To simulate the plant model in discrete time, a Zero-Order Hold (ZOH) conversion is applied to the continuous transfer function $G_\theta(s)$. The discrete

transfer function $G_\theta(z)$ is computed using MATLAB's `d2c` command with a sample time of $T = 0.05$ s. With ZOH, conversion to s-domain is:

$$G_s(s) = \frac{b}{s+a}, \quad a = -\frac{\ln a_1}{h}, \quad b = b_1 \frac{1-a_1}{h\, a_1}.$$

Substituting $\hat{a}_1, \hat{b}_1$ and $h = 0.05\,\mathrm{s}$ gives

$$\boxed{G_s(s) = \frac{15.8686}{s + 17.4054}}.$$

Reflection:
The OLS approach yielded a good fit of 66% shown in figure Figure 2 to the observed servo behavior and supports the hypothesis of dominant first-order dynamics. However, real servos often exhibit saturation, dead zones, or hysteresis not captured by this model. These will try to be identified in part's b and c.

Algorithm 1 OLS Identification of the Servo Transfer Function $G_s(z)$

Require: Time series $\{\alpha(k), u(k)\}_{k=0}^{N}$
Build regressor matrix $\mathbf{\Phi} = [\,\alpha(k-1) \;\; u(k-1)\,]$
Build output vector $\mathbf{Y} = \alpha(k)$
Solve normal equations $\hat{\boldsymbol{\theta}} = (\mathbf{\Phi}^\top \mathbf{\Phi})^{-1} \mathbf{\Phi}^\top \mathbf{Y}$
Extract $\hat{a}_1, \hat{b}_1$ and form $G_s(z) = \dfrac{\hat{b}_1 z^{-1}}{1 - \hat{a}_1 z^{-1}}$

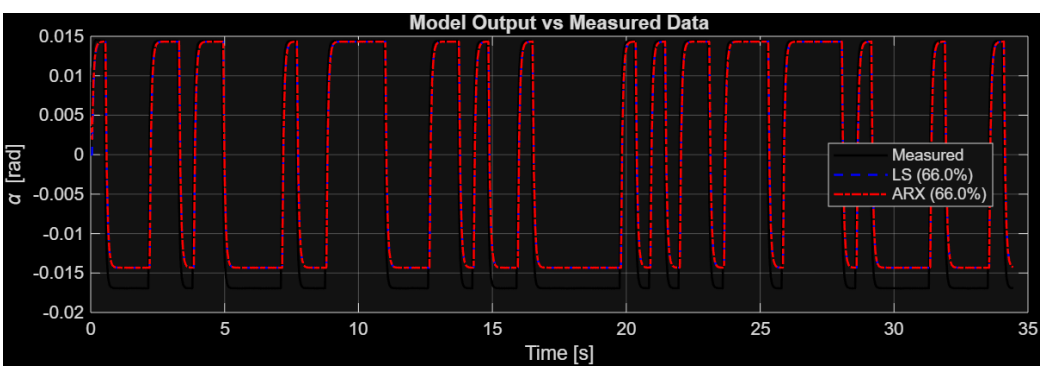

Figure 2: **Part A.** Measured vs. modelled servo angle, ARX and Ordinary Least Squares Modeled, with a fit of 66% compared to the measured data

4 Part B – Coupling Gain c and Plant $G_\theta(s)$

Summary:
This section estimates the physical coupling between servo deflection and beam rotation . A second-order model is derived from first principles, and the coupling gain c is identified via finite-difference regression using experimental data. This completes the open-loop model from input u to output θ.

4.1 Finite-Difference Regression

Second backward difference approximates $\ddot{\theta}$:

$$\ddot{\theta}(t_k) \approx \frac{\theta(k) - 2\theta(k-1) + \theta(k-2)}{h^2} = \frac{c}{2}\big[u(k-1) + u(k-2)\big].$$

Re-arranged into the LS (Least Squares) form $S\,c = Y$ with $Y(k) = \theta(k) - 2\theta(k-1) + \theta(k-2)$ and $S(k) = \frac{h^2}{2}\big[u(k-1) + u(k-2)\big]$, the scalar estimate is simply

$$\boxed{\hat{c} = \frac{S^\top Y}{S^\top S} = 29.1376} \tag{B1}$$

4.2 Composite Transfer Function

Cascading $G_s(s)$ from Part A with equation (2) gives

$$G_\theta(s) = \frac{\hat{c}\,15.8686}{s^2(s + 17.4054)} = \frac{461.8310}{s^3 + 17.3900s^2},$$

When the ZOH of $G_\theta(s)$ is applied:

$$G_\theta(z) = \frac{0.0078 + 0.0256z^{-1} + 0.0051z^{-2}}{1 - 2.4192z^{-1} + 1.8383z^{-2} - 0.4192z^{-3}}.$$

5 Part C – Recursive Least-Squares Tracking of the Coupling Gain c

Summary:
To detect time-varying dynamics or environmental effects, a scalar Recursive Least Squares (RLS) estimator was implemented that tracks the coupling

gain c over time. Unlike the batch estimate in Part B, RLS provides a method for updating c with each new sample, improving adaptability in variable conditions.

5.1 Motivation

The rigid-body relation $\ddot{\theta}(t) = \frac{c}{I_b}\,\alpha(t)$ suggests that the single gain c may drift due to unmodeled dynamics or environmental changes. Instead of re-identifying the full model, we estimate c using RLS; all other plant coefficients are kept from Parts A–B. This greatly reduces complexity while still detecting mechanical drift.

5.2 Regressor Definition

With sample time $T = 0.05\,\mathrm{s}$ and backward-difference approximation

$$\ddot{\theta}(kT) \;\approx\; \frac{\theta(k) - 2\theta(k-1) + \theta(k-2)}{T^2},$$

the discrete identification equation becomes

$$\underbrace{\theta(k) - 2\theta(k-1) + \theta(k-2)}_{Y(k)} \;=\; \underbrace{\frac{T^2}{2}\big[\alpha(k-1) + \alpha(k-2)\big]}_{S(k)}\, c + \,\varepsilon(k).$$

For each new sample k we update the RLS coefficients:

$$e(k) = Y(k) - S(k)\,\hat{c}(k-1), \tag{3}$$

$$R(k) = \lambda R(k-1) + S^2(k), \tag{4}$$

$$\hat{c}(k) = \hat{c}(k-1) + \frac{S(k)}{R(k)}\,e(k), \tag{5}$$

where $R(k)$ is the scalar covariance and $\lambda \in (0, 1]$ is the forgetting factor. $\lambda \in (0, 1]$ defines how much the new coefficients influence the system. Here we set $\lambda = 1$ (no forgetting) to emulate batch LS while still yielding a running estimate.

5.3 Results on Five Independent Trials

Applying the RLS equation((5)) to each of the five experimental data sets gives the following converged values:

Dataset	$\hat{c}_{\text{final}}$	R_{final}
1	30.5187	8.6091×10^{-7}
2	30.5436	8.6909×10^{-7}
3	26.8859	8.7719×10^{-7}
4	27.3392	9.6327×10^{-7}
5	30.9104	8.7849×10^{-7}

Reflection:

The RLS estimator converged quickly on most trials and agreed closely with the batch estimate from Part B, validating its correctness. Trials 3 and 4 revealed slightly lower gain values, which may indicate reduced thrust or unmodeled dynamics. Although we used $\lambda = 1$ for stability, tuning this parameter could enhance responsiveness to abrupt changes in future implementations.